MG Week 4: Assignment 4.1: Solution System of Interest: Home Weather Station

April 2, 2026

1 Solution: System of Interest (SoI)

Objective: Building the logical architecture of the SoI.

In this phase of specifying the solution domain, you should imagine yourself as the systems architect, working to deliver the logical architecture of the SoI. The essential goal of building this model is to define the logical subsystems of the SoI, which leads to the identification of work packages. As mentioned before, the logical architecture of the SoI can be stored in a separate model, apart from the problem domain definition. The highlighted area in the following figure shows exactly which part of the solution domain we are discussing.

As you may conclude by looking at the design layout of the MagicGrid framework (see the highlighted area in the following figure), the system-level solution domain model specifies the system requirements, structure, behavior, and parameters. Additionally, every SysML element you create in the logical system architecture model must satisfy at least one system requirement. Otherwise, there is no motivation for having it in the logical system architecture model. Therefore, you have to start and end the system-level solution domain model with system requirements: first, specify them and, when you have the entire logical system architecture model, establish traceability relationships between them and logical system architecture elements [PHD21].

			Pillar				
Domain			Requirements	Structure	Behavior	Parameters	Safety & Reliability
	Problem	Black Box	Stakeholder Needs	System Context	Use Cases	Measures of Effectiveness (MoEs)	Conceptual and Functional Failure Mode & Effects Analysis (FMEA)
		White Box		Conceptual Subsystems	Functional Analysis	MoEs for Subsystems	Conceptual Subsystems FMEA
	Solution		System Requirements	System Structure	System Behavior	System Parameters	System Safety & Reliability (S&R)
			Subsystem Requirements	Subsystem Structure	Subsystem Behavior	Subsystem Parameters	Subsystem S&R
			Component Requirements	Component Structure	Component Behavior	Component Parameters	Component S&R
	Implementation		Implementation Requirements				

Figure 1: MagicGrid® System Solution

Use the MagicGrid BOK as a reference.

Use the following equation for your WiFi range calculation.

$distance(km) = 10^{(T-S+G-L-M-32.44-20*log(f))/20}$

T - transmit power of the Raspbeery Pi in dBm. You control this in your design. Typical range 9.8 - 28.2 dBm

S - sensitivity of the monitoring device in dBm. Assume -80

G - gain in dBm, assume 0 as typical value for wifi device without external antenna

L - loss in dBm, assume 0 for ideal link budget, fade margin will be used to account for environmental uncertainties.

M - fade margin in dBm, accounts for unknowns. Use 30 dBm for high fading such as in an urban environment. Use 12 dBm for low fading in rural open environment

The value 32.44 accounts for physical contants and unit conversions. This equation is only valid for frequency in MHz.

f - RF frequency in MHz, use 2412 MHz.

Tasks:

1. MBSE Planning
2. System Requirements
3. System Structure
4. System Behavior
5. System Parameters
6. Traceability to Requirements
7. Review

Outputs:

1. System requirements in REQ
2. Requirement trace to Stakeholder needs in Matrix
3. System and subsystems in BDD
4. Logical to conceptual subsystems trace in Matrix
5. System/subsystem interfaces in BDD
6. System states in STM
7. MoEs and constraints in BDD
8. Calculation in PAR
   (a) Wifi range calculation
9. Early verification results stored in model
10. Review captured in model

2 Solution: Subsystem Requirements and Structure

Note: you are only doing the first half of the logical architecture in week 4.

Build the logical architecture of the subsystems.

A complete logical subsystem architecture (LSSA) model consists of the specification of subsystem requirements, structure, behavior, and parameters. Additionally, every SysML element you create in this model must satisfy at least one subsystem requirement; otherwise, there is no motivation for having it in the LSSA model. Therefore, you have to start and end the LSSA model with subsystem requirements: first specify them and, when your model is complete,

establish traceability relationships between them and SysML elements [PHD21].

			Pillar				
			Requirements	Structure	Behavior	Parameters	Safety & Reliability
Domain	Problem	Black Box	Stakeholder Needs	System Context	Use Cases	Measures of Effectiveness (MoEs)	Conceptual and Functional Failure Mode & Effects Analysis (FMEA)
		White Box		Conceptual Subsystems	Functional Analysis	MoEs for Subsystems	Conceptual Subsystems FMEA
	Solution		System Requirements	System Structure	System Behavior	System Parameters	System Safety & Reliability (S&R)
			Subsystem Requirements	Subsystem Structure	Subsystem Behavior	Subsystem Parameters	Subsystem S&R
			Component Requirements	Component Structure	Component Behavior	Component Parameters	Component S&R
	Implementation		Implementation Requirements				

Figure 2: MagicGrid® Subsystem Solution

Use the MagicGrid BOK as a reference.

Tasks:

1. MBSE Planning
2. Subsystem Requirements
3. Subsystem Structure
4. Team Peer Review

Outputs:

1. Subsystem requirements in REQ
2. Subsystem to system deriveReqt and refine trace in Matrix
3. Subsystem architecture in BDD
4. Components and interfaces in BDD
5. Review captured in model

References

[PHD21] Aurelijus Morkevičius PH.D. *MagicGrid® BOOK OF KNOWLEDGE*. A Practical Guide to Systems Modeling using MagicGrid from Dassault Systèmes. VITAE LITERA, 2021. ISBN: 978-609-454-554-2.